AMERICAN INTERNATIONAL UNIVERSITY- BANGLADESH

FACULTY OF SCIENCE AND TECHNOLGY

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EXP 6: To determine the refractive index of the material of a prism using a spectrometer

1.Which one is the Snell's law?

(a) $\sin(i)/\sin(r) = n_2/n_1$ (b) $\sin(r)/\sin(i) = n_2/n_1$

(c) $\sin(i)/\cos(r) = n_2/n_1$ (d) $\sin(r)/\sin(i) = n_1/n_2$

Ans: (a)

2. Which eyepiece is used in the telescope of a spectrometer?

(a) Newton's eyepiece (b) Einstein eyepiece

(c) Snell's eyepiece (d) Ramsden's eyepiece

Ans: (d)

3.What kind of lamp you often use in the laboratory?

(a) mercury vapor lamp (b) sodium lamp

(c) metal Halide lamp (d) Fluorescent lamp

Ans: (b)

4.Which light is the mixture of lights of different color?

(a) Green light (b) Purple light

(c) White light (d) yellow light

Ans: (c)

5.The speed of light on each face is related to the equation of,

(a) $C_1/C_2 = n_1/n_2$ (b)$C_1/C_2 = n_2/n_1$

(c) $C_2/C_1 = n_2/n_1$ (d)$C_2/C_1 = n_2/n_1$

Ans: (b)

6. To determine the refractive index of the material of a prism you need?

(a) Calipers (b) speedometers

(c) Spectrometers (d) thermometers

Ans: (c)

7.Which color varies the angle of minimum and maximum deviation?

(a) green and white (b) red and purple

(c) yellow and violet (d) red and violet

Ans: (d)

8.The refractive index of the material of the prism can be calculated by the equation,

(a) $n = \sin((A+D)/2)/\sin(A/2)$ (b) $n = \cos((A+D)/2)/\cos(A/2)$

(c) $n = \mathrm{Sin}(A/2)/\sin((A+D)/2)$ (d) $n = \cos(A/2)/\cos((A+D)/2)$

Ans: (a)

9.Dispersive power of a prism can be found with sodium light

(a) True (b) False

Ans: (b)

10. On what factors, the dispersive power depends?

(a) power and material (b) power and wavelength

(c) material and wavelength (d) none of the above

Ans: (c)

Ex-7: To verify the laws of transverse vibration of strings and to determine frequency of a turning Fork by Melde's Experiment

11. Among the three laws of transverse vibration what is the law of length?

(a) $f \propto \frac{1}{\sqrt{m}}$ (b) $f \propto \frac{1}{L}$ (c) $f \propto T$ (d) None

Ans: (b)

12. What is the mode in which a string can vibrate?

(a) Longitudinal (b) Transverse (c) None (d) Both a and b

Ans: (d)

13. In a transverse wave the distance between a crest and trough is equal to:

(a) λ (b) $\frac{\lambda}{2}$ (c) 2λ (d) $\frac{\lambda}{4}$

Ans: (d)

14. When transverse wave is reflected on going from a denser to rarer medium then at the boundary the reflected wave undergoes a phase change of_____.

(a) $0°$ (b) $90°$ (c) $180°$ (d) $-90°$

Ans: (a)

15.The oscillation of a transverse wave is_____to the direction of motion

(a) Parallel (b) Perpendicular (c) Both (d) None

Ans: (b)

16. Two vibrating springs of the same material but lengths L and 2L have radii 2r and r respectively. They are stretched under the same tension. Both the strings vibrate in their fundamental modes, one of the lengths L with frequency f_1 and the other with frequency f_2. The ration is given by?

(a) 2 (b) 4 (c) 8 (d) 1

Ans: (d)

17. A transverse wave is transporting energy from east to west. The particles of the medium will move?

(a) East to west (b) North to west

(c) Both eastward and westward (d) Both northward and southward

Ans: (d)

18. A travelling wave in a stretched string is described by the equation $y = A\sin(kx-\omega t)$. The maximum particle velocity is _____.

(a) $A\omega$ (b) ω/k (c) $d\omega/dk$ (d) x/t

Ans: (a)

19. A tunning fork has frequency 288 Hz. Find the number of vibration of tuning when the sound from tuning fork reaches 330m in air. Velocity of sound in air is 330 m/s. Number of vibration done by tuning fork is_____.

(a) 388 (b) 480 (c) 288 (d) 188

Ans: (C)

20. In a resonance tube with tuning fork of frequency 512Hz, first resonance occurs at water level equal to 30.3cm and second resonance occurs at

63.7cm. The maximum possible error in the speed of sound is?

(a) 51.2 cm/s (b) 102.4m/s (c) 204.8cm/s (d) 1536cm/s

Ans: (a)

Ex-8: Radius of curvature of Plano convex lens using newton's ring

21.When light traveling in air gets reflected from water surface,there is_____

a) phase change of π

b) phase change of π/2

c) phase change of π/4

d) no phase change

Ans: (a)

22.phase difference of 2π corresponds to a path difference of

a) λ/2

b) λ

c) λ/4

d) 3λ/2

Ans: (b)

23.For light of wavelength 6000Å,which of the following path difference will give constructive interference?

a)0.1 μm

b)0.3 μm

c)0.6 μm

d)0.5 μm

Ans: (c)

24.The condition for obtaining secondary maxima in the diffraction pattern due to single slit is :

a) $a\sin\theta = n\lambda/2$

b) $a\sin\theta = (2n-1)\lambda/2$

c) $a\sin\theta = n\lambda$

d) $a\sin\theta = (2n-1)\lambda$

Ans: (b)

25.In the propagation of polarised light waves, the angle between the plane of vibration and the plane of polarization is

a)0°

b)45°

c)90°

d)180°

Ans: (c)

26. What are the uses of Newton's rings?

a. To determine of light (monochromatic).

b. To determine of a liquid.

c. To measure the radius of spherical surface.

d. To measure expansion coefficient of crystal.

Ans: (a)

27. What are the essential conditions for observing the interference phenomenon in the laboratory?

a. The two sources should be coherent.

b. The two sources must emit waves of same wavelength and time period.

c. The sources should be monochromatic.

d. The amplitudes of the interfering waves should be equal or nearly equal.

Ans: (a)

28. The radius of half period zones are proportional to

a. 1/√n

b. √n

c. n -3/2

d. n 3/2

Ans: (b)

29. Diffraction of light is observed when the size of the obstacle is

a. very large

b. very small

c. howsoever large or small

d. comparable with the wavelength of light

Ans: (d)

30. The sources of light are said to be coherent if the waves produced by them have the same:

a. Wavelength

b. Amplitude

c. Wavelength and a constant phase difference

d. Amplitude and the same wavelength

Ans: (c)